Mario Lavina

GitHub LinkedIn

Education

Imperial College London — **London, UK**
MSc. Artificial Intelligence — 2021–2022

London School of Economics — **London, UK**
BSc. Mathematics with Economics — 2018–2021

Experience

FINESSE — **Los Angeles, CA**
Engineering Manager, ML Engineer — Jan 2023–Present

Lead engineering at a $300M Series A startup (20 headcount, $16M revenue). Streamlined team from 8 to 5 while doubling release cadence. Built CI/CD, code review, and sprint processes from scratch.

- Mapped **16 manual workflows** with the CEO and two department heads and automated all in **three months**, saving about **60 hours a week**, measured across logistics, CX and executive teams: scripts for the 11 that needed no language model, **three AI agents** (**Python**, **Claude API**, 3 shared **MCP servers**) for the other 5.
- Built two internal dashboards: a **Next.js** operations view (launched September 2026) joining our design platform, logistics systems and **Stripe Payouts**, so staff see each creator design, its estimated China-to-US arrival and payout in one place, and a monitor for every agent and automation I built.
- Built the versioned API (**FastAPI**, Postgres on AWS) our AI agents call for every order, stock and refund action. It enforces each agent's limits in code, not in the prompt: the support agent took **0 unauthorized actions in 1,000 attack cases**, including instructions hidden in warehouse and courier replies, and **0 in production** (3 weeks).
- Built a customer-support agent that handles **90% of 3,000 monthly tickets** (from 17,000 monthly orders) with no staff, **75%** also with no repeat contact in 7 days; the support lead's audit of 200 a month against courier records finds **96% right**. In **70% of 600 monthly delivery problems** it gets the trace, reship or refund from the warehouse or courier itself.
- Replayed 3 years of support chats, split by date: agent matched the human on **94% of 1,200 held-out recent cases**, per a grader validated on 200 hand-labelled ones. In older chats, adding each chat's date cut wrong-warehouse errors from **12% to <1%**.
- Put refund authority in code, not the prompt: each staff approval, rejection or edit of the agent's proposal becomes a test case, and a refund type turns automatic after 300 straight approvals. Four have, cutting tickets that need staff from **20% to 10%** in a year; finance traced a random 300 of a year's 2,100 automatic refunds to courier records: **4 unwarranted, against 9 of 300 for staff**.
- Interviewed **12 of the 400 creators** whose designs we manufacture; most asked when a design reaches US shoppers and when they get paid. Gave our support agent both answers from factory, freight and payout data: **310 of the 400** used it in month one, and status emails fell from **120 to 10 a week**.
- Set the team's rules for coding agents, which now write most of our code: a written spec per task, type checks and integration tests in CI, human review, and no agent edits to payments code. Taught **6 non-engineer colleagues** to turn agent logs into test cases, now **a third of the suite**. With the team cut from 8 to 5, releases went from **every 2 weeks to weekly** and production bugs fell from **10 to 2 a month**.
- Built AI-powered QA system using Claude agents with Figma design verification and Metro simulator — reviews **40 PRs/week**, replaced 2 QA engineers (**$160K/yr**), cut release cycles in half, reduced production bugs by **83%**.
- Stood up the data program behind a garment-image-to-tech-pack agent: wrote a 40-field spec and rejection rules with the model team, ran a 100-style pilot, then had 6 technical designers capture **3,000 styles and 10,000 fabrics** in 10 weeks, with weekly usable-rate reporting (**88% to 97%**). For 300 products, design to first sample fell from **6 weeks to 4 days**.
- Built demand forecasting (XGBoost + SageMaker) that triggers reorders across 24K SKUs and 30 suppliers, cutting overproduction **30%** and saving **$1M/yr**; planning the weekly China-to-US shipment from it moved 35% of volume from air to sea, cutting freight cost per unit **21%** with stockouts flat.

- Purchasing agent, built with the production team for 30 suppliers in China (1,200 orders a year), awards each order across three quotes on projected 60-day profit and confirms ship dates twice before departure: on-time departure rose from **66% to 88%**; **9% more profit** than lowest-price buying on 400 replayed orders.
- Shipping agent settles stock shortfalls with the packing warehouse in Chinese, closing **95% of 140 disputes** within a day with the stock found or credited; pre-orders ship **10 minutes** after stock arrives (was 2 days).
- Shipped mobile app in 3 months after inheriting a stalled project (React Native), grew it from $0 to **$300K/mo revenue — 10K WAU**, 6× higher conversion than web (4.2% vs 0.7%), now driving 23% of company revenue at 16% higher AOV. Built and tested its **TypeScript** API (Express, TypeORM, Postgres) with Testcontainers integration tests in CI.
- Redesigned and ran 100+ A/B tests on conversion-optimized features, increasing conversion rates by **0.9–2.0%** and doubling annual revenue from $6M to $12M.
- Shipped a text-to-image model (**Qwen-Image**) serving **40K generations/day** at 4s latency under 60 concurrent requests, adopted by 72% of WAU; users who created and posted AI-generated designs showed **2× D30 retention**.
- Built the labeling tool for a **1M-image** training set: a model pre-fills each label and labelers confirm or fix it with keyboard shortcuts, cutting time per image from **40s to 15s** while agreement on a 2,000-image double-labeled check held at **94%**, saving **6,900 labeler hours**.
- Trained a caption model on **CLIP** image features with 8,000 stylist-written captions; in a blind test on 500 fashion images, stylists preferred its captions over GPT-4V's **64%** of the time, and its tags filled missing color, fabric and fit fields on **40,000 product pages**.
- Built a Ray pipeline that ingests **100K+ TikTok and Instagram posts a day**, extracts category, purchase intent and sentiment from their comments, and feeds a real-time engine ranking fashion trends by likes, comments, sentiment and author, with an average rank error of **5** in validation.
- Built the **KPI dashboard** the CEO reviews **each Monday** (revenue, margin, cash, stock cover, web and app KPIs), refreshed nightly.
- Coordinated with the recruiter for engineering roles: **800 applicants**, interviews booked within a day, a **decision within a day** of the final interview; **3 hires** in a year.
- Closed our third-party warehouse in New Jersey: sold **167,233 of 188,840 units** at a **27% margin** in 10 weeks, donated the rest, and ended the contract 2 months early, saving **$100K in storage per month**.
- Proposed and ran a 2-day, no-downtime AWS-to-Google Cloud move that cut the cloud bill from **$10K to $1.7K a month**: deleted unused resources, tuned the database onto a smaller server and archived old data.

Imperial Driverless **London, UK**
Lead ML Engineer Sep 2021–Jul 2022

- Led a 5-engineer Motion Planning team to a **top-5 finish** in Imperial Driverless's first-ever competition, with **80% run completion**; owned path-and-motion planning on a Jetson Xavier race car at up to 30 mph.
- Built a **real-time SLAM** system (ORB-SLAM2 + occupancy mapping), hitting sub-50ms pose latency and <5cm drift / km in GPS-denied environments.
- Implemented LiDAR-camera sensor-fusion module (**ICP + EKF**, Velodyne VLP-16 + camera), cutting track-boundary localization error **4x (0.5m →0.12m)** and tightening the racing-line planning tolerance.
- Shipped a fused perception stack - (**YOLOv3**) cone detection + (**MiDaS**) monocular depth - sustaining 37 FPS on Jetson within the car's power budget, feed the planner at control-loop rate.

Research

Generating Safety Critical Scenarios, *Dr Dario Paccagnan and Dr Panagiotis Angeloudis*

- Developed RL framework (MADDPG, DQN) synthesizing edge-case accident scenarios for autonomous vehicles, achieving **7% higher collision coverage** than existing scenario generators. Distributed training across 16×A100 GPUs using Ray RLlib with **4× faster convergence**.

Selected Projects

Auto Parts Finder, *VibeCon 2026 hackathon by Emergent (175 of 2,000 applicants selected), team of 2* Oct 2026

- Built an **AI voice agent** (ElevenLabs) that phones car dealerships for mechanics: from a VIN and part, it gets price and stock in a **median 2.5 min**, against 9 min for 4 LA mechanics (60 timed calls, 1 in 3 unanswered). A second voice agent answers dealers' return calls, matched to the request by caller number.

- In **30 live calls** to 12 dealers it got **17 quotes, 16 at the exact price** (checked against the recordings), and a repair shop ran 14 real requests through it in a day, getting 9 quotes. Tuned on 50 simulated edge cases (43/50 to **50/50**), it scored **7/8** on 8 of the 20 real dealer calls, held out from tuning.

Skills

Programming Languages: Python, SQL, Mojo (experimenting)
Frameworks: PyTorch, Tensorflow, OpenCV, NLTK, HuggingFace (transformers, diffusers), RLlib, OpenAI Gym, Ray
Software/Tools: AWS Stack (S3, EC2, RDS, Sagemaker, Cloudfront, Quicksight), Pinecone, W&B, FastAPI, Render